

EDUCATION

NORTHWESTERN UNIVERSITY

Data Science (Artificial Intelligence Specialization)

M.S. | Expected Dec 2026 | Current GPA: 4.0 / 4.0

UNIVERSITY OF CALIFORNIA LOS ANGELES

Economics

B.A | Jul 2017

PROFESSIONAL EXPERIENCE

Los Angeles Capital Management LLC., Jan 2022 - current

Lead Analytics Engineer & Global Trading Associate

- Designed and operated real-time trading execution and analytics systems processing millions of order execution records per day, enabling liquidity-aware decision-making and institutional portfolio execution strategies.
- Led end-to-end design of integrated trading and AI systems spanning execution infrastructure, canonical data platforms, and agentic analytics workflows in production environments.

Data Engineering & Trading Data Platforms

Spark, Databricks, Snowflake, SQL

- Architected distributed analytics and ETL pipelines processing high-volume execution and liquidity datasets for transaction cost analysis and trading intelligence workflows.
- Serving as committee lead for implementation of medallion architecture standards to improve data lineage, governance, scalability, and downstream AI integration capabilities.
- Directed migration of core financial data infrastructure (security master + market data systems) improving scalability, performance, and data interoperability.
- Implemented version-controlled using Git-based repositories, code review practices, and governed deployment processes to improve maintainability, auditability, and production reliability.
- Built reusable transformation and data quality frameworks to support secure, scalable, and extensible enterprise analytics pipelines across multiple business domains.
- Delivered and engineered centralized cloud-native analytics platforms integrating Databricks, Snowflake, and distributed reporting workflows, reducing report generation times by 50%.
- Developed secure cloud-native Bloomberg API integration infrastructure enabling scalable enterprise market data access within internal environments.

Applied Data Systems & Machine Learning Infrastructure

Python, Claude, Mosaic AI, Databricks

- Built LLM-powered analytics interfaces enabling natural language querying over structured trading and execution datasets.

- Designed agent-based analytics workflows for automated summarization and structured reporting of trading activity.
- Implemented standardized LLM-to-data system interface layer enabling controlled, auditable access to enterprise datasets.
- Collaborated directly with trading, compliance, IT, and portfolio management teams to productionize AI workflows within live enterprise infrastructure environments.
- Integrated AI-driven analytics workflows into production trading systems used by portfolio managers and trading stakeholders.

Quantitative Systems & Execution Analytics

Python, SQL, Monte Carlo Simulation.

- Developed Monte Carlo-based optimization system for trade prioritization and liquidity-sensitive execution strategies.
- Built real-time decision-support pipelines combining execution data, market data, and analytics features for live trading environments.
- Performed portfolio performance attribution, return decomposition, and risk analytics using SQL and Python.
- Delivered an analytics tool which is supporting live trading workflows and time-sensitive decision-making environments, increasing trade list completion by 3%.

Los Angeles Capital Management LLC., Aug 2017 – Jan 2022

Performance Measurement Analyst

- Designed and maintained scalable SQL-based ETL pipelines supporting portfolio performance analytics.
- Automated reconciliation and reporting processes across large-scale financial datasets and benchmark performance systems.
- Conducted quantitative analysis on portfolio returns, benchmark attribution, and operational trading metrics using SQL.

LANGUAGES

Python, PySpark, SQL, R, VBA, C++, Java.

AI / CONVERSATIONAL SYSTEMS

RAG Systems, Claude AI, Databricks Mosaic AI, Agentic Workflows, NLP, Vector Search, Semantic Retrieval, LLM Orchestration, Monte Carlo Simulation, TensorFlow.

CLOUD & INFRASTRUCTURE

Databricks, Spark, Snowflake, Docker, CI/CD, ETL Pipelines, Medallion Architecture, Data Governance.

DATABASE & SEARCH

PostgreSQL, MySQL, Milvus, Elasticsearch, Vector Databases.

VISUALIZATION & ENTERPRISE TOOLS

Tableau, Bloomberg API, CRD, Eze, Triton.